Bayesian Volatility at Scale

> Loading project_proposal.pdf… [OK]
> Parsing Magris & Iosifidis (2023)… [OK]
> Executing VI_GARCH.ipynb… [DONE]
> QML · MCMC · BBVI · Full-cov VI… [OK]

Runtime: NumPy + PyTorch | Data: MSFT | Method: VI

The project in plain English

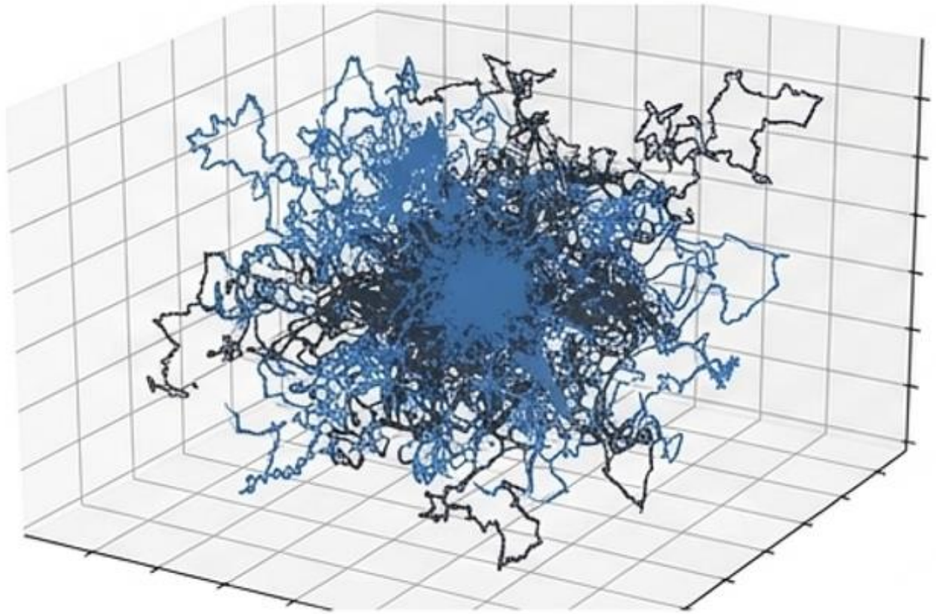
What is volatility?	Why forecast it?	What did we compare?	What did we find?
How bumpy a stock's ride is. Stormy days cluster together, a bit like the weather.	Banks and investors use it to set risk limits, price options and decide how much to invest.	The trusted-but-slower Bayesian method (MCMC) against a machine-learning shortcut (VI) for fitting the standard volatility model, GARCH.	Same forecasts (within 1%). Simple VI is over-confident; full VI is not. For this small model, the shortcut is not faster.

Analogy: MCMC explores a landscape on foot, step by step. VI flies over it and fits a simple map. The question is whether the map is accurate enough to navigate by.

Data: Microsoft daily prices 2018–2023 (the paper's own example), plus 28 large US companies.

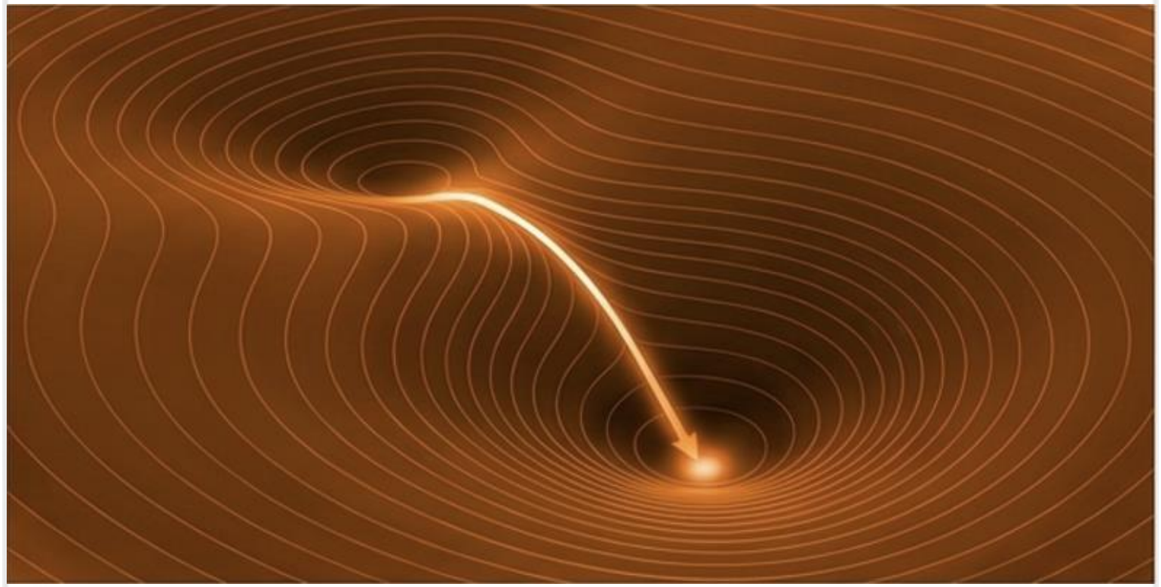
MCMC is the Bayesian gold standard, but it becomes a bottleneck at scale

Traditional Bayesian Estimation (Markov Chain Monte Carlo)



- **Method**: Random-walk posterior sampling
- **Cost**: Sequential draws, tuning, diagnostics
- **Scalability**: Mixes slowly in high dimensions

The Chosen Architecture (Variational Inference)



- **Method**: Optimize the ELBO (momentum / Adam)
- **Cost**: One convergence curve, generic tuning
- **Scalability**: Built for high-dimensional models

Variational Inference recasts Bayesian estimation into a scalable optimization problem

1. The Intractable True Posterior



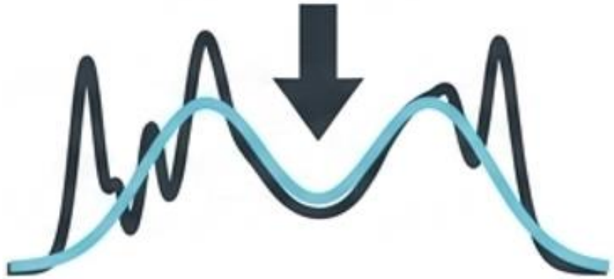
Marginal likelihood **p(y)** is often intractable.

2. The Variational Family



A family of tractable proxy distributions $\boldsymbol{q}(\boldsymbol{\theta})$ parameterized by $\boldsymbol{\zeta}$.

3. The Optimization Bridge

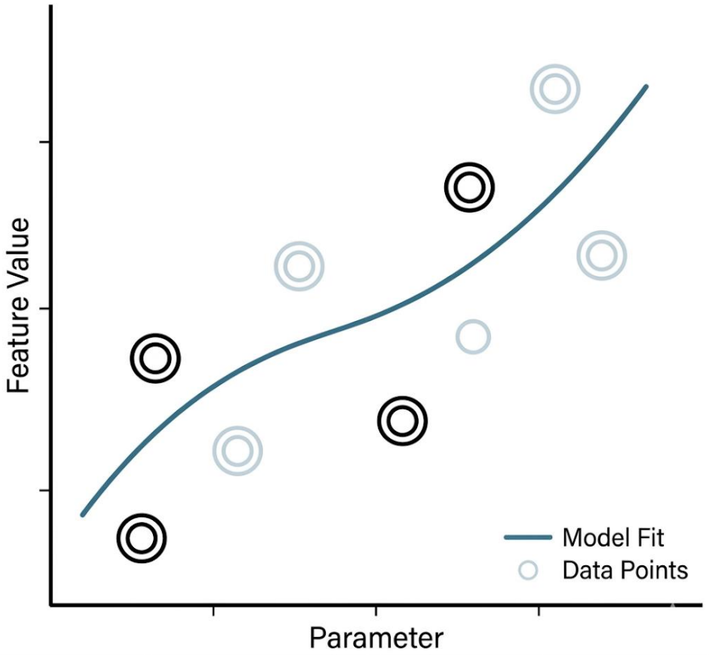


Minimize the Kullback-Leibler (KL) Divergence to force the proxy $\boldsymbol{q}(\boldsymbol{\theta})$ to match the true posterior.

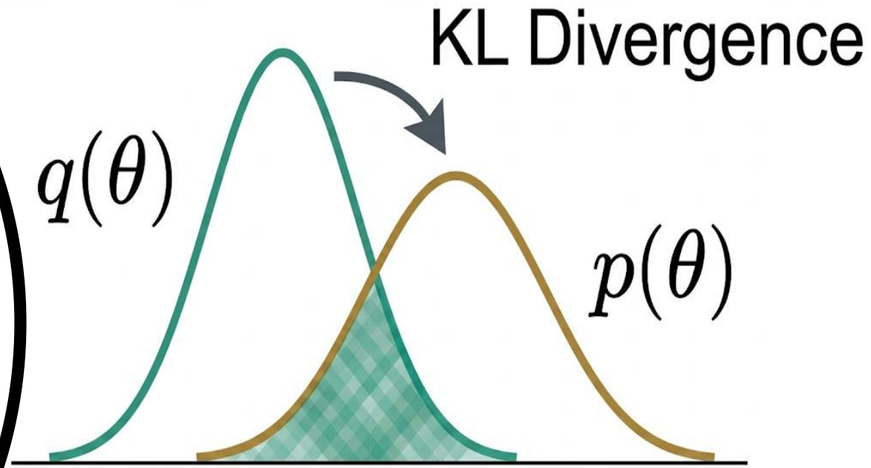
In plain English: instead of exploring every possible answer, VI fits the closest simple bell curve, like tuning a dial until the fit stops improving.

The Evidence Lower Bound (ELBO) objective balances data fit against prior beliefs

The Expected Log-Likelihood



$$ELBO = \mathbb{E}[log\ p(data \mid \theta)] - \mathrm{KL}(q(\theta) \,||\, p(\theta))$$



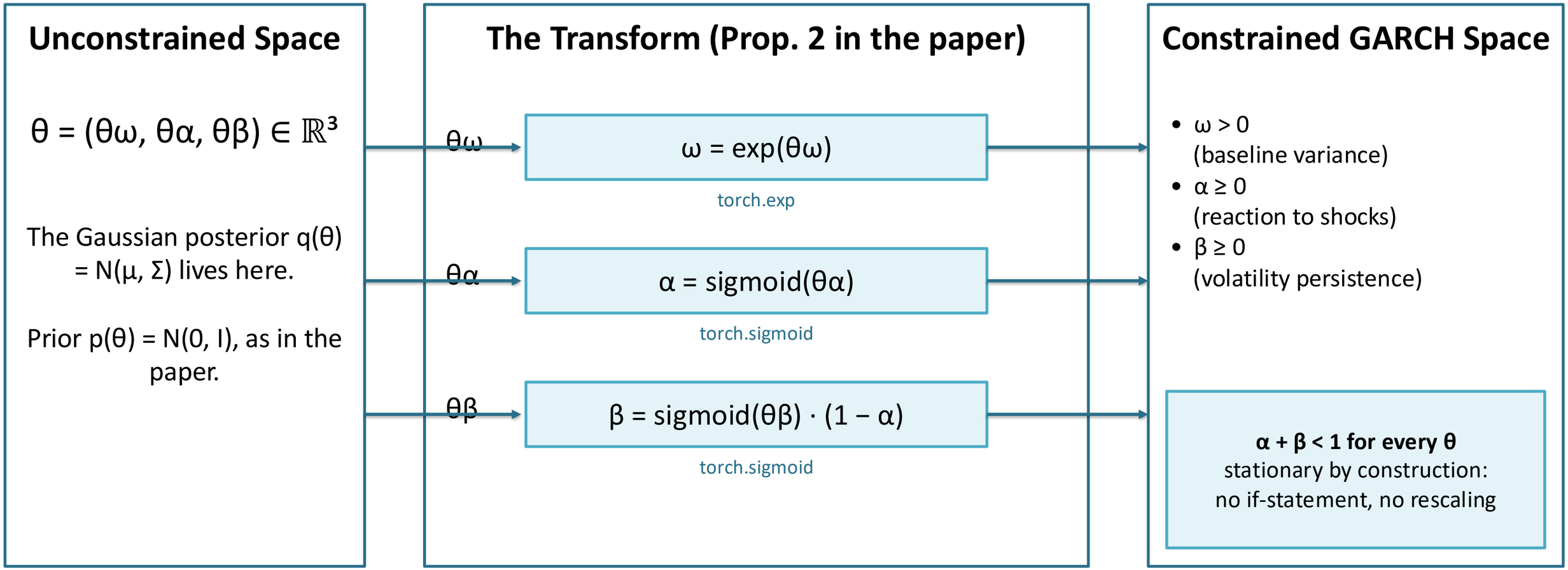
The KL Divergence

Maximizes the model's fit to the empirical financial data.

Averaged over random guesses.

Penalizes the variational distribution $\boldsymbol{q(\theta)}$ for deviating too far from standard prior $\boldsymbol{p(\theta)}$**.** Acts as a natural regularizer

Paper transforms map unconstrained VI parameters into a valid, stationary GARCH



In plain English: the three dials get built-in guardrails, so the model can never produce an impossible forecast (negative or exploding volatility).
Fix vs. first draft: replaces the old "if α + β ≥ 0.99, rescale" rule, which was not smooth and not the paper's method.

The VI training loop: sample, score, update in PyTorch

1. Sample Parameters



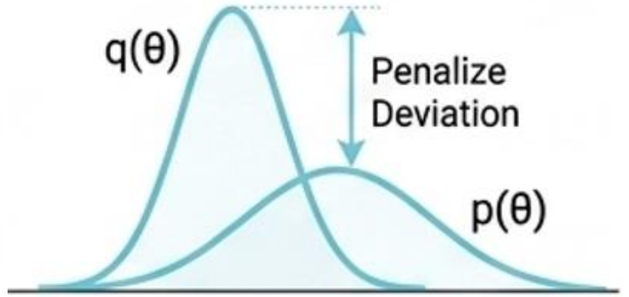
$$q(\theta) = N(\mu, \sigma^2)$$

2. Calculate Expected Log-Likelihood



Compute GARCH Likelihood over MC samples.

3. Compute KL Divergence



$$KL(q(\theta) \,||\, \rho(\theta))$$

4. Backpropagate ELBO Loss

loss.backward() → Adam step

Autodiff gradients

5. Update Parameters

Adam Optimizer Step (lr = 0.01)

Update μ, σ^2

The ELBO Objective

$$ELBO = \mathbb{E}[log\, p(data \mid \theta)] - \mathrm{KL}(q(\theta) \,||\, p(\theta))$$

In plain English

Guess the dials, score the fit, nudge the guess, repeat about 1,500 times until the score stops improving.

Two versions

BBVI (paper): needs only fit scores; assumes the dials are unrelated.

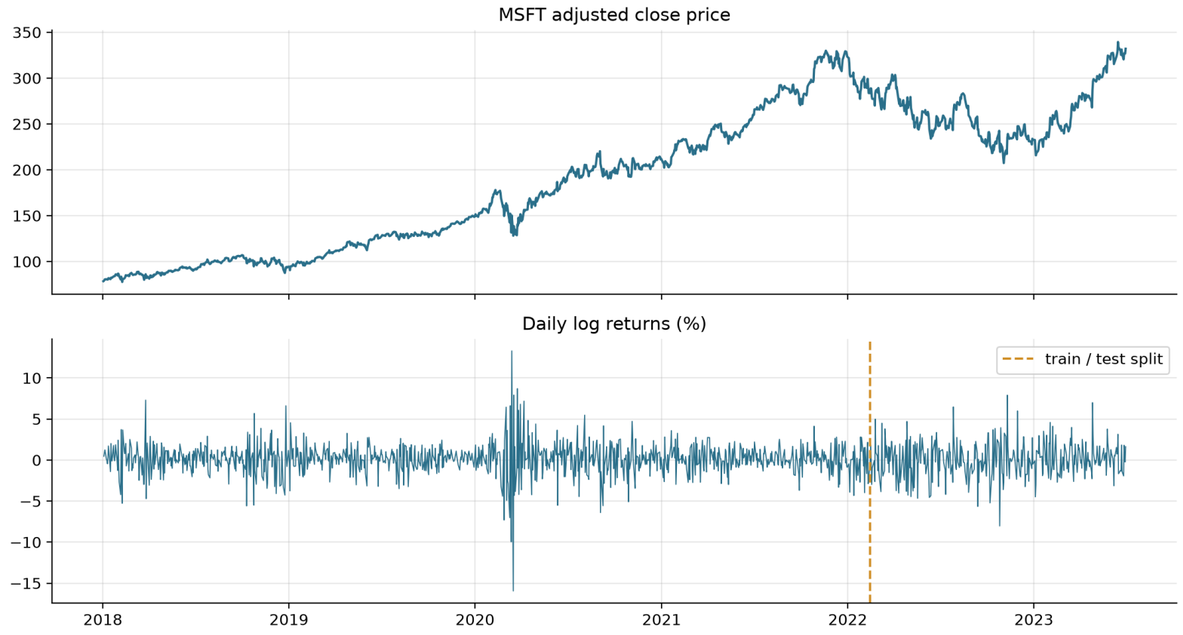
Full-cov VI: PyTorch does the calculus; captures how the dials move together.

Data pipeline: the paper's Microsoft case study, cleaned and split in time order

Data Pipeline

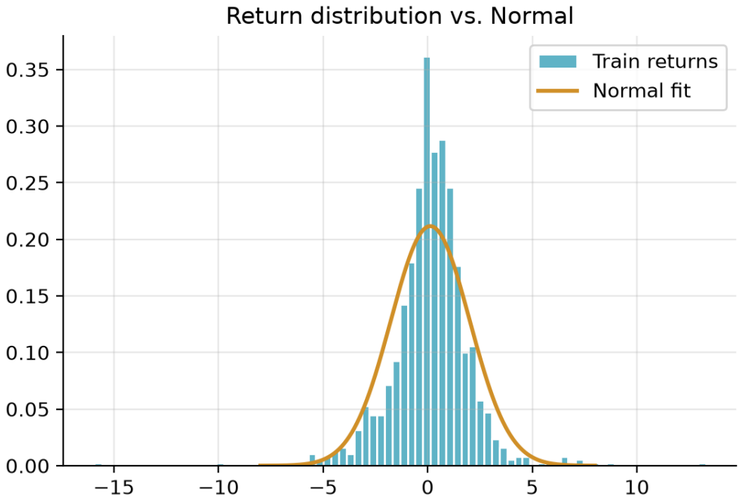
Step	Detail
Source	Yahoo Finance (yfinance), adjusted close
Asset	Microsoft (MSFT), as in the paper; plus a Dow 30 panel
Window	2 Jan 2018 – 30 Jun 2023: 1,383 days → 1,382 returns
Cleaning	0 missing, 0 duplicates, 0 non-positive prices; longest gap 4 days (holidays)
Returns	Daily % change: $r_t = 100 \cdot (\log P_t - \log P_{t-1})$
Split	Learn on first 75% (to 11 Feb 2022); test on last 25% (the "exam")

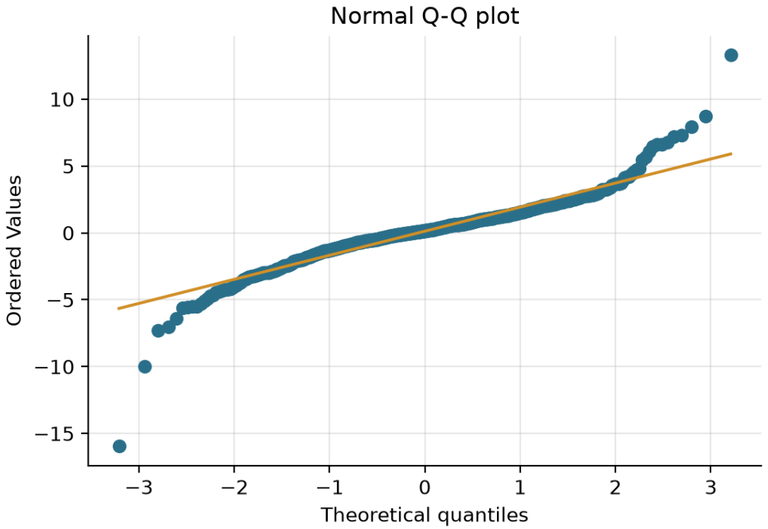
Prices and Returns

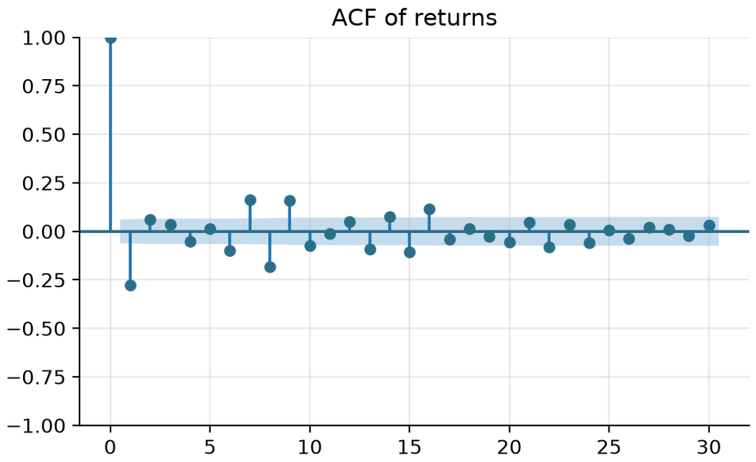


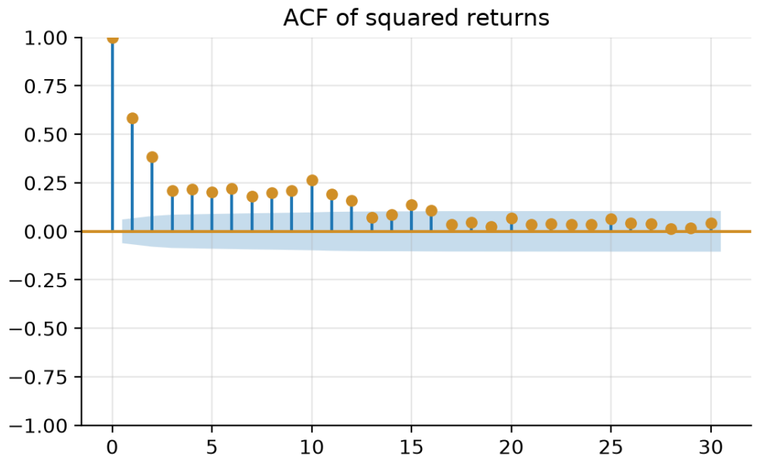
Takeaway: the test period (2022 rate hikes) is more volatile than most of the training period, so it is a genuine out-of-sample challenge.

Exploratory analysis: MSFT returns show the stylised facts GARCH is built for









Fat tails
Extreme days are far more common than a bell curve predicts (excess kurtosis 9.5; Jarque-Bera $p < 0.001$)

Volatility clustering
Stormy days follow stormy days (Ljung-Box on r^2: 890, $p < 0.001$)

Predictable turbulence
Yesterday's swings predict today's (ARCH-LM: 393, $p < 0.001$)

Average return ≈ 0
The daily drift (0.12%) is tiny next to the daily swings (1.9%), so we model only the swings

In plain English: turbulence comes in waves and crashes are more common than "normal" suggests; GARCH handles the waves, Student-t the crashes.

Our GARCH(1,1) estimates replicate the paper's Table II for Microsoft

Method	ω (ours)	α (ours)	β (ours)	ω (paper)	α (paper)	β (paper)
QML	0.160	0.186	0.769	0.15	0.19	0.77
MCMC	0.217 ± 0.049	0.203 ± 0.031	0.731 ± 0.036	0.200 ± 0.046	0.203 ± 0.031	0.737 ± 0.036
BBVI (diagonal)	0.214 ± 0.022	0.201 ± 0.023	0.732 ± 0.023	0.196 ± 0.021	0.199 ± 0.023	0.740 ± 0.023
Full-cov VI*	0.220 ± 0.050	0.200 ± 0.029	0.732 ± 0.035	0.205 ± 0.047	0.201 ± 0.030	0.736 ± 0.035

ω = baseline turbulence · **α** = reaction to yesterday's shock · **β** = memory of past turbulence. Bayesian columns: average ± spread of plausible values. *Paper column = MGVB, its full-covariance VI.

Replicated

Our numbers match the published paper (all within ≈ 0.02), so the code is doing the right thing.

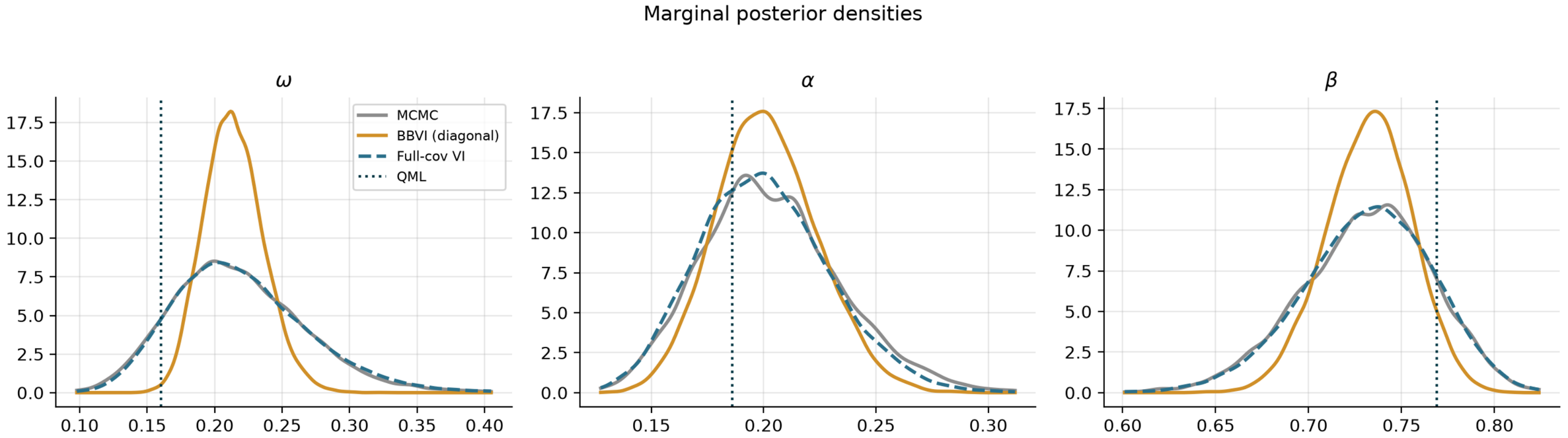
Persistence

$\alpha + \beta \approx 0.955$: a volatility shock has a half-life of ≈ 15 trading days.

Out-of-sample check

Test NLL (QML): 764.6 ours vs 761.4 in the paper.

Full VI gets the uncertainty right; the simpler diagonal VI is over-confident



Spread (uncertainty) of ω

MCMC: 0.049
Full-cov VI: 0.050
BBVI: 0.022 (−55%)

Spread (uncertainty) of β

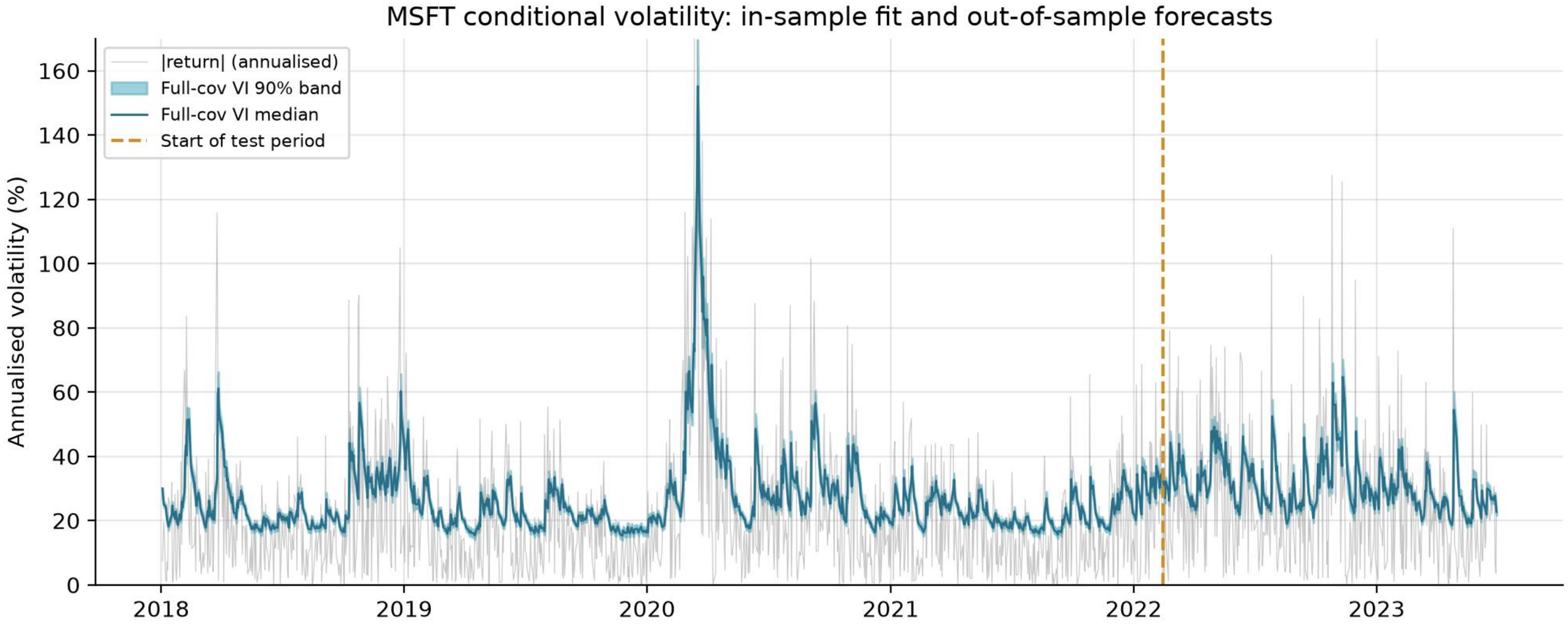
MCMC: 0.036
Full-cov VI: 0.035
BBVI: 0.023 (−35%)

Fit score, ELBO (higher = better)

Full-cov VI: −1945.3
BBVI: −1946.2
Same ordering as the paper

In plain English: all methods agree on the best guess, but simple VI claims to be about twice as sure as it should be, like a forecaster who never says "maybe".

Bayesian estimation turns volatility into a forecast with credible bands



In plain English: like a weather forecast with a range: the line is expected turbulence, the shaded band its likely range. It catches the COVID-19 crash and the 2022 sell-off, which it never saw while learning.

Average 90% band width (test): MCMC 3.5 · Full-cov VI 3.5 · BBVI 3.1 vol-points (diagonal VI is over-confident).

VI forecasts as well as QML and MCMC, but it is not faster for a 3-parameter GARCH

Out-of-sample accuracy: % difference vs QML

Method	NLL	RMSE	MAD	QLIK
MCMC	+0.15	+0.23	−0.60	+0.26
BBVI (diagonal)	+0.19	+0.18	−0.82	+0.32
Full-cov VI	+0.16	+0.13	−0.94	+0.27

NLL, RMSE, MAD, QLIK are forecast-error scores (lower = better). Every gap is under 1% (paper: sub-1% on average); negative = better than QML.

Computational cost (MSFT, one run)

Method	Run time	Likelihood evals
QML	< 0.01 s	≈ 380
MCMC (RWM)	≈ 0.3 s	30,000
BBVI (diagonal)	≈ 1 s	125,000
Full-cov VI (PyTorch)	≈ 16 s	15,000

In plain English: all methods forecast equally well. For this small 3-dial model, the trusted method (MCMC) is actually quicker, so VI is not a speed win here.

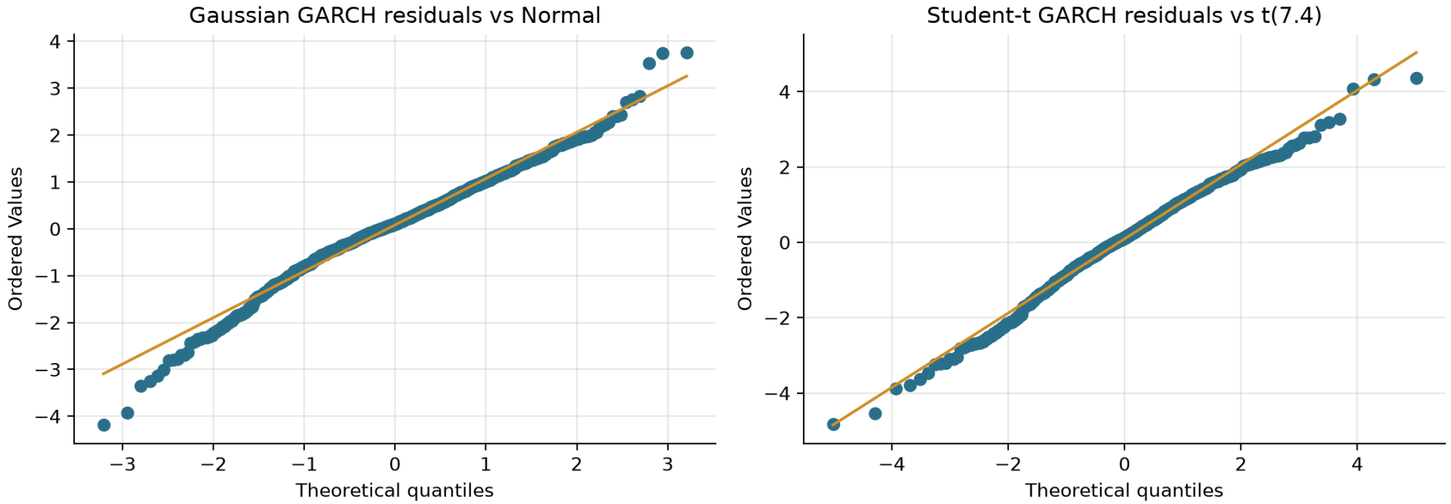
VI's real advantages: one ELBO curve to monitor, the same hyperparameters for every stock, a reusable analytical posterior, and scalability to high-dimensional models where MCMC mixes slowly.

Extension: Student-t innovations capture the fat tails that Gaussian GARCH misses

Heavy Tails (Student-t Innovations)

- **The problem:** Gaussian-GARCH residuals are still fat-tailed (left Q-Q).
- **The extension:** $\nu = 2 + \exp(\theta_\nu)$, estimated jointly with full-covariance VI (4 parameters).
- **The result:** $\nu \approx 7.4$ (90% interval 5.2–10.2), far from Normal.

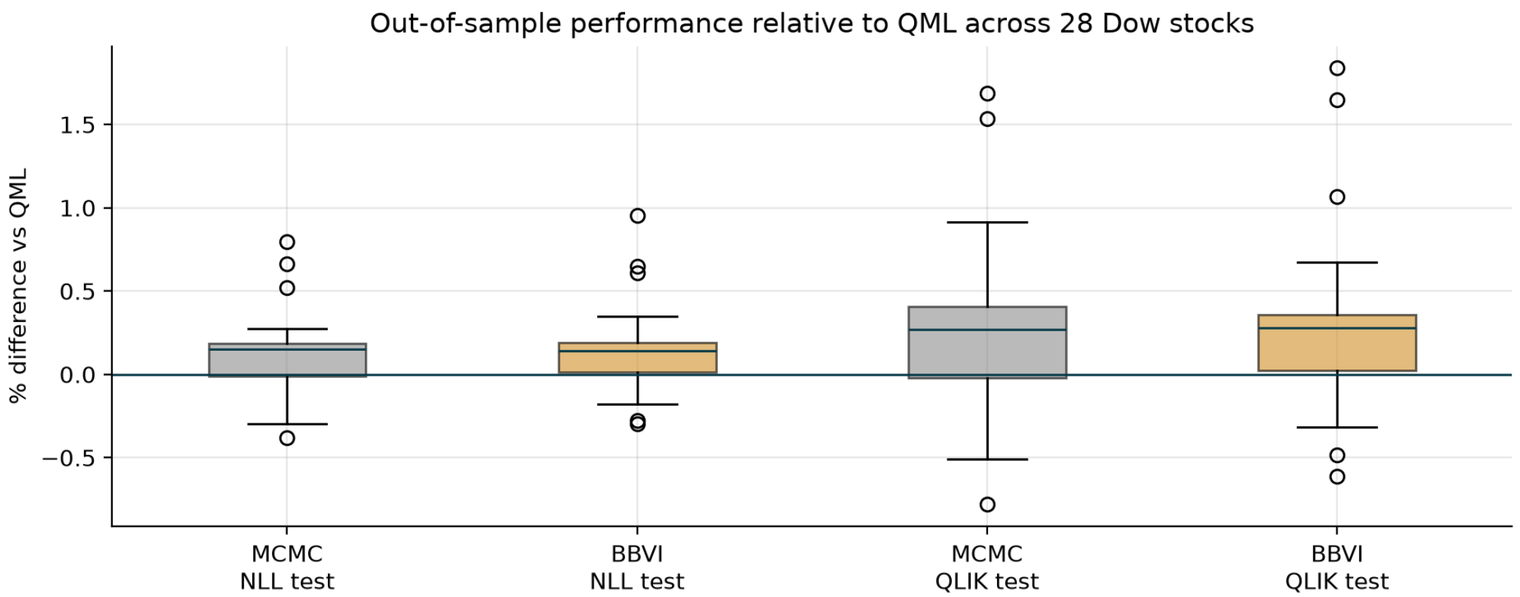
In plain English: this version expects the occasional crash day, so real markets surprise it less.



Criterion	Gaussian VI-GARCH	Student-t VI-GARCH
Fit score, ELBO (higher = better)	−1945.3	**−1934.6 (+10.7)**
Forecast score on unseen data	−765.9	**−754.1 (+11.8)**
Volatility path	almost identical	almost identical

Scaling up: the same conclusion holds across 28 Dow Jones stocks

Data: 28 Dow 30 stocks (DOW, WBA dropped: incomplete) | 2018–2023 | **Split:** 75% / 25% | identical hyperparameters for every stock



Mean % deviation vs QML

Test NLL: MCMC +0.12 · BBVI +0.14
Test QLIK: MCMC +0.25 · BBVI +0.28
Train NLL: QML lowest for 100% of stocks (it maximises it by design)

Average time per stock: MCMC ≈ 0.3 s, BBVI ≈ 1.1 s

This project (28 stocks)
MCMC and BBVI forecasts sit within ≈ 0.3% of QML on average and are almost indistinguishable from each other.

Magris & Iosifidis (488 stocks)
Across seven GARCH-family models and four VI optimisers, deviations from QML are sub-1% on average (paper, Table I).

Execution Summary & Core Takeaways

Replication

- **Goal**: Implement VI for GARCH as in Magris & Iosifidis (2023).
- **Result**: QML, MCMC and BBVI match the paper's Table II within ≈ 0.02; the transforms guarantee $\alpha + \beta < 1$.

Accuracy & Uncertainty

- **Goal**: Match MCMC/QML accuracy.
- **Result**: VI is within 1% of QML out of sample (MSFT and 28 stocks). Full-cov VI ≈ MCMC posterior; diagonal BBVI is over-confident.

Fat Tails & Cost

- **Goal**: Model real-world returns efficiently.
- **Result**: Student-t ($\nu \approx 7$) wins on ELBO and test log-likelihood. At 3 parameters MCMC is cheap; VI pays off through generic tuning and scale.

Variational Inference is an accurate, well-calibrated alternative to MCMC for GARCH, as long as the variational family is rich enough (full covariance).

Next steps: GJR / EGARCH / FIGARCH · natural-gradient optimisers (QBVI, MGVB) · realised-volatility proxies from intraday data (e.g. WRDS TAQ)

Jargon buster

Term	What it means
Volatility	How big the daily price swings are (shown annualised, in %)
GARCH(1,1)	Forecasts tomorrow's volatility from today's shock (α) and today's volatility (β), plus a baseline (ω)
Stationary (α + β < 1)	Shocks eventually fade, so volatility cannot explode
QML	Classic method that finds the single best-fitting setting of the dials
Prior / Posterior	What we believe about the dials before / after seeing the data
MCMC	Traditional Bayesian method: a long random walk through plausible settings
Variational Inference (VI)	Machine-learning shortcut: fit the closest simple bell curve to the posterior
ELBO	The fit score VI maximises (higher = better); also used to compare models
BBVI vs full-covariance VI	Diagonal VI assumes the dials are unrelated; full VI captures how they move together
Train / test	Data the model learns from / unseen "exam" data used to check forecasts
NLL, QLIK, RMSE, MAD	Forecast-error scores: lower is better
Student-t / fat tails	Allows extreme days to be more common than a bell curve predicts

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Patton, A. J. (2011). Volatility Forecast Comparison Using Imperfect Volatility Proxies. Journal of Econometrics, 160(1), 246–256.
Code & data: github.com/SGhuman123/Fordham-Time-Series-Econometrics (VI_GARCH.ipynb)